Chapter 13 — Halogens

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Chlorine at room temperature: (a) Greenish-yellow gas (b) Red-brown liquid (c) Grey solid (d) Colourless liquid

2. Bromine at room temperature: (a) Red-brown liquid (b) Grey solid (c) Gas (d) Colourless gas

3. Iodine at room temperature: (a) Gas (b) White powder (c) Greyish-black solid (d) Liquid

4. Volatility from Cl_2 to I_2: (a) Increases (b) Same (c) Zero (d) Decreases

5. Strongest oxidizing halogen: (a) Br_2 (b) Cl_2 (c) F_2 (d) I_2

6. Chlorine displaces bromine from: (a) Water only (b) KBr solution (c) KCl solution (d) NaF solution

7. H_2 + Cl_2 needs: (a) Darkness (b) No initiator (c) Sunlight (d) Low pressure only

8. Main disinfecting species: (a) O_2 (b) Cl^- (c) Na^+ (d) HOCl

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why does iodine have a higher boiling point than bromine?

2. Why is fluorine a stronger oxidizing agent than chlorine?

3. How can chlorine displace bromine from potassium bromide?

4. Why is the F_2 bond unexpectedly weak?

5. How does HOCl kill bacteria?

6. How does pH affect chlorine disinfection?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain trends in volatility, bond strength and oxidizing power of halogens. **(4)**

(b) Describe reactions of halogens with hydrogen. **(4)**

Q.4 (a) Explain the use of chlorine in water purification. **(4)**

(b) Describe factors affecting chlorine disinfection. **(4)**

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